

EXP.NO: 6. Write a Servlet to demonstrate session tracking using HttpSession.

AIM:

To demonstrate session tracking using HttpSession in a Servlet-based web application. The exercise involves implementing a simple login system where a user can log in with their username and password, and their session will be tracked throughout their interaction with the web application.

ALGORITHM:

1. Input:

- The user will enter their username and password in a login form.
- The servlet will validate the credentials and create an HTTP session for the user if authentication is successful.

2. Steps for Implementing Session Tracking with HttpSession:

1. Create the Login Form (HTML):

- o Create an HTML form with two input fields for username and password.
- o Provide a submit button to send the data to the servlet for authentication.

2. Create the Login Servlet:

- o Write a Java servlet class to handle login requests.
- o The servlet will:
 - Retrieve the username and password from the form.
 - Authenticate the user by comparing the credentials to a stored value (e.g., hardcoded or from a database).
 - Create a new HTTP session for the user if the credentials are valid.
 - Set session attributes such as the user's username to track the session.

3. Validate User Credentials:

- o In the servlet, check if the username and password match the expected values.

- o If valid, create a session and store the username in the session.

- o If invalid, redirect the user back to the login page with an error message.

4. Create and Track User Session:

- o Upon successful login, use the HttpSession object to store the session data.

- o Use the session.setAttribute("username", username) method to save the username.

- o Optionally, set session timeout using

session.setMaxInactiveInterval(time_in_seconds) to manage session expiration.

5. Redirect to a Protected Page:

- o After a successful login, the user will be redirected to a protected page (e.g., a dashboard or home page).

- o The protected page should check if the user is authenticated by checking the session attribute.

6. Logout and Session Invalidation:

- o Provide a logout feature that invalidates the session using session.invalidate() when the user logs out.

- o After logout, redirect the user back to the login page.

7. Session Expiration:

- o If the session expires or the user manually logs out, the session will be invalidated, and the user will be redirected to the login page.

Code:

```
package com.demo;
```

```
import jakarta.servlet.*;

import jakarta.servlet.http.*;

import jakarta.servlet.annotation.*;

import java.io.*;

@WebServlet("/DashboardServlet")

public class DashboardServlet extends HttpServlet {

    protected void doGet(HttpServletRequest request, HttpServletResponse response)
        throws ServletException, IOException {

        HttpSession session = request.getSession(false);

        if(session == null){
            response.sendRedirect("login.html");
            return;
        }

        String user = (String) session.getAttribute("user");
        String loginTime = (String) session.getAttribute("loginTime");

        PrintWriter out = response.getWriter();

        out.println("<html><body style='font-family:Arial'>");
```

```
out.println("<h2>Welcome " + user + "</h2>");

out.println("<p><b>Login Time:</b> " + loginTime + "</p>");

out.println("<p>Session is active</p>");


out.println("<a href='LogoutServlet'>Logout</a>");


out.println("</body></html>");


}

}

<!DOCTYPE html>

<html>

<head>

<title>Login Page</title>

<link rel="stylesheet" href="style.css">

</head>

<body>

<div class="login-box">
```

```
<h2>Login</h2>
```

```
<form action="LoginServlet" method="post">
```

```
<input type="text" name="username" placeholder="Enter Username" required>
```

```
<input type="password" name="password" placeholder="Enter Password" required>
```

```
<button type="submit">Login</button>
```

```
</form>
```

```
</div>
```

```
</body>
```

```
</html>
```

```
package com.demo;
```

```
import jakarta.servlet.*;
```

```
import jakarta.servlet.http.*;
```

```
import jakarta.servlet.annotation.*;
```

```
import java.io.*;
```

```
import java.time.LocalDateTime;
```

```
import java.time.format.DateTimeFormatter;
```

```
@WebServlet("/LoginServlet")

public class LoginServlet extends HttpServlet {

    protected void doPost(HttpServletRequest request, HttpServletResponse response)
        throws ServletException, IOException {

        String username = request.getParameter("username");
        String password = request.getParameter("password");

        if(username.equals("admin") && password.equals("1234")){

            HttpSession session = request.getSession();

            session.setAttribute("user", username);

            String loginTime = LocalDateTime.now()
                .format(DateTimeFormatter.ofPattern("dd-MM-yyyy HH:mm:ss"));

            session.setAttribute("loginTime", loginTime);

            response.sendRedirect("DashboardServlet");

        }else{

            response.getWriter().println("<h3>Invalid Login</h3>");
```

```
}
```

```
}
```

```
}package com.demo;
```

```
import jakarta.servlet.*;
```

```
import jakarta.servlet.http.*;
```

```
import jakarta.servlet.annotation.*;
```

```
import java.io.*;
```

```
@WebServlet("/LogoutServlet")
```

```
public class LogoutServlet extends HttpServlet {
```

```
protected void doGet(HttpServletRequest request, HttpServletResponse response)
```

```
throws ServletException, IOException {
```

```
HttpSession session = request.getSession(false);
```

```
if(session != null){
```

```
session.invalidate();
```

```
}
```

```
response.sendRedirect("login.html");
```

```
}
```

```
}
```

```
body{
```

```
font-family: Arial;
```

```
background: linear-gradient(135deg,#667eea,#764ba2);
```

```
height:100vh;
```

```
display:flex;
```

```
justify-content:center;
```

```
align-items:center;
```

```
}
```

```
.login-box{
```

```
background:white;
```

```
padding:40px;
```

```
width:320px;
```

```
border-radius:10px;
```

```
box-shadow:0px 0px 15px rgba(0,0,0,0.2);
```

```
text-align:center;
```

```
}
```

```
.login-box h2{
```

```
margin-bottom:20px;
```

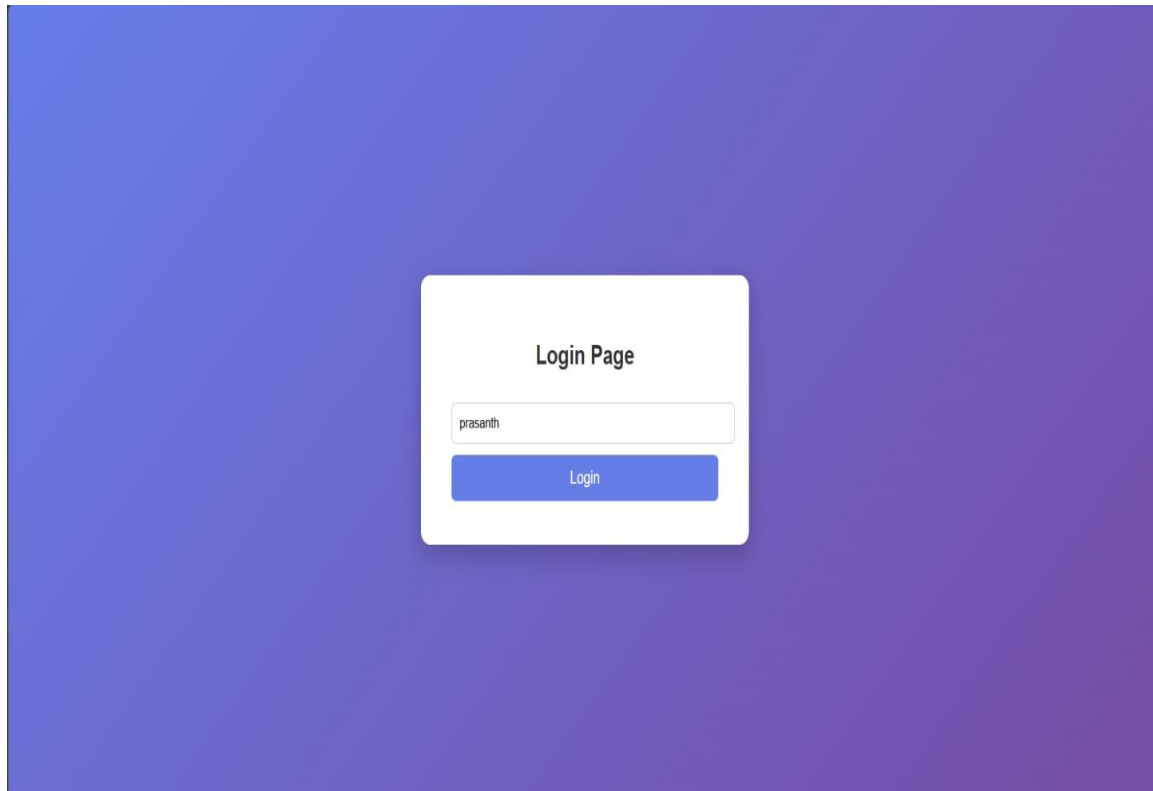
```
}
```

```
input{  
width:100%;  
padding:10px;  
margin:10px 0;  
border-radius:5px;  
border:1px solid #ccc;  
}
```

```
button{  
width:100%;  
padding:10px;  
background:#667eea;  
border:none;  
color:white;  
font-size:16px;  
border-radius:5px;  
cursor:pointer;  
}
```

```
button:hover{  
background:#5a67d8;  
}
```

Output:



RESULT:

A simple login system is created where the user's session is tracked by a servlet to demonstrate session tracking using HttpSession.